Insight: Financial Report Analysis Engine - Progress Report

Group259
Wang Yixin 72512080
Zhang Jiajun 72510615
Wang Hongyuan 72510222

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1 Project Background and Objectives

With the rapid development of financial markets, the number and complexity of financial research reports are increasing. Analysts and investors need to quickly extract key information from a large number of reports to make informed investment decisions. However, traditional manual reading methods are inefficient and difficult to handle massive amounts of information.

In recent years, the development of large language models (LLM) and retrieval-augmented generation (RAG) technology has provided new solutions for financial report analysis. By combining the capabilities of vector retrieval and large language models, intelligent analysis and question-answering for financial reports can be achieved.

The core objective of the Insight: Financial Report Analysis Engine project is to build a vertical domain RAG question-answering system that benchmarks the NotebookLM experience, focusing on the analysis of long, information-dense financial PDF reports, and providing page-level citation tracking.

Specific objectives include:

1. Implement efficient parsing and processing of financial PDF reports
2. Build high-quality vector indexes to support fast and accurate information retrieval
3. Integrate local large language models to provide intelligent question-answering and analysis capabilities
4. Implement page-level citation tracking to enhance the credibility of answers
5. Establish a comprehensive evaluation system to ensure system performance

2 Project Technology Stack

The project adopts the following technology stack:

- **Frontend Framework**: Gradio
- **Backend Framework**: LlamaIndex v0.10+
- **Vector Database**: ChromaDB

- **Embedding Model**: BAAI/bge-m3
- **Large Language Model**: Qwen2.5-7B/72B (locally deployed via Ollama)
- **PDF Parsing**: PyMuPDF
- **Evaluation Framework**: RAGAS

3 Completed Work

3.1 System Architecture Design

The overall system architecture design has been completed, adopting a highly cohesive, low-coupling modular design, consisting of four core components:

- Data ingestion layer: Responsible for PDF parsing and chunking
- Retrieval and generation layer: Implements vector retrieval and hybrid retrieval, as well as answer generation
- Evaluation closed-loop layer: Implements RAGAS evaluation
- Interaction layer: Builds Gradio frontend

3.2 Core Module Implementation

3.2.1 Data Ingestion Layer

Implemented PDF report parsing and chunking functionality, supporting semantic chunking to improve the accuracy of information retrieval.

3.2.2 Retrieval and Generation Layer

Implemented vector retrieval and hybrid retrieval functionality, integrated RAPTOR tree-based summarization technology, improving the efficiency and accuracy of retrieval.

3.2.3 Evaluation Closed-loop Layer

Integrated RAGAS evaluation framework, implementing comprehensive system performance evaluation.

3.2.4 Interaction Layer

Built Gradio frontend, providing a user-friendly interactive interface.

3.3 Environment Configuration

Completed system environment configuration, successfully deployed and running on Mac Studio (M3 Ultra).

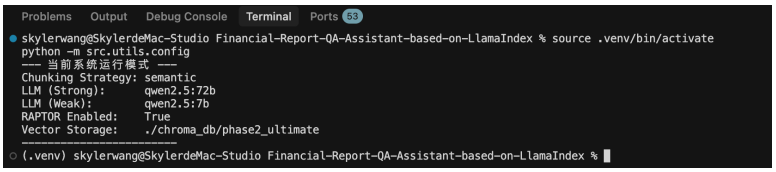

Figure 1: System Environment

Figure 2: Advanced Architecture (Semantic + RAPTOR)

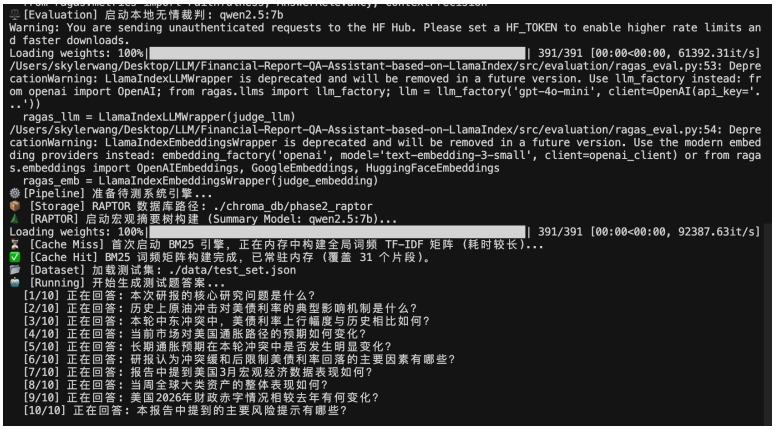

Figure 3: Advanced Evaluation

4 Ongoing Work

4.1 Performance Optimization

Currently optimizing system performance, including:

- Optimizing vector index building and retrieval speed
- Improving chunking strategy to enhance information retrieval accuracy
- Optimizing large language model inference speed

4.2 Feature Enhancement

Currently enhancing system features, including:

- Improving the accuracy of page-level citation tracking
- Adding multilingual support
- Enhancing understanding of tables and charts

4.3 User Experience Improvement

Currently improving user experience, including:

- Optimizing frontend interface response speed
- Adding more interactive features
- Providing more detailed result explanations

5 Challenges and Solutions

5.1 Challenge 1: PDF Parsing Accuracy

Financial reports often contain complex formats, such as tables and charts, which traditional PDF parsing tools struggle to extract accurately.

Solution: Adopted PyMuPDF for PDF parsing, combined with custom post-processing logic to improve parsing accuracy.

5.2 Challenge 2: Long Text Processing

Financial reports are typically long, and traditional RAG methods struggle to effectively handle long text.

Solution: Adopted semantic chunking and RAPTOR tree-based summarization technology to decompose long text into manageable chunks and build hierarchical summary structures.

5.3 Challenge 3: Comprehensive Evaluation

How to comprehensively evaluate RAG system performance is a challenge.

Solution: Integrated RAGAS evaluation framework to evaluate system performance from multiple dimensions, including answer relevance, faithfulness, and completeness.

6 Next Steps (Two-week Plan)

1. Complete performance optimization work to improve system response speed
2. Complete feature enhancement work to add table and chart understanding capabilities
3. Optimize UI design and user experience
4. Deploy the system to production environment
5. Collect user feedback to further improve the system

6. Write detailed technical documentation
7. Conduct comprehensive system testing and evaluation

7 Conclusion

The Insight: Financial Report Analysis Engine project has made significant progress, completing system architecture design and core module implementation. By adopting advanced RAG technologies, the system can effectively analyze financial PDF reports and provide accurate question-answering and analysis capabilities.

Although some challenges were encountered during the project, they were effectively addressed through team efforts and technological innovation. In the future, we will continue to optimize system performance, enhance system features, and improve user experience to provide a more powerful tool for financial report analysis.